

# **SRM VALLIAMMAI ENGINEERING COLLEGE**

**(An Autonomous Institution)  
SRM Nagar, Kattankulathur – 603 203**

**DEPARTMENT OF COMPUTER APPLICATIONS**

**QUESTION BANK**



**III SEMESTER**

**PMC303 - DATA SECURITY AND PRIVACY**

**Regulation – 2024**

**Academic Year 2025 – 2026(Odd Semester)**

*Prepared by*

**P.Pandi Deepa**

**Assistant Professor**

**Department of Computer Applications**

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DEPARTMENT OF COMPUTER APPLICATIONS

## QUESTION BANK

**SUBJECT : PMC303 - DATA SECURITY AND PRIVACY**

**SEM/YEAR: III / II**

### **UNIT - I: ATTACKS AND PRIVACY**

Attacks: Analyzing common attack vectors – Data Security – Probabilistic reasoning about attacks –Data security mitigations. Privacy aware Machine learning and Data Science: Privacy preserving techniques in ML - Open-source libraries for PPML Architecting privacy in Data and ML projects

### **UNIT - I: PART – A**

| <b>Q. No</b> | <b>Question</b>                                                                     | <b>BT Level</b> | <b>Competence</b> | <b>Course Outcome</b> |
|--------------|-------------------------------------------------------------------------------------|-----------------|-------------------|-----------------------|
| 1            | Define <b>attack vector</b> with an example.                                        | BTL-1           | Remember          | CO1                   |
| 2            | What is a <b>brute-force attack</b> in cybersecurity?                               | BTL-1           | Remember          | CO1                   |
| 3            | List any two types of <b>social engineering attacks</b> .                           | BTL-1           | Remember          | CO1                   |
| 4            | What is the difference between <b>passive</b> and <b>active attacks</b> ?           | BTL-1           | Remember          | CO1                   |
| 5            | Define <b>data confidentiality</b> and <b>data integrity</b> .                      | BTL-1           | Remember          | CO1                   |
| 6            | What do you mean by <b>probabilistic reasoning</b> in the context of attacks?       | BTL-1           | Remember          | CO1                   |
| 7            | Give any two examples of <b>data security breaches</b> .                            | BTL-1           | Remember          | CO1                   |
| 8            | What is a <b>zero-day attack</b> ?                                                  | BTL-1           | Remember          | CO1                   |
| 9            | Explain how a <b>phishing attack</b> works with a real-time example.                | BTL-2           | Understand        | CO1                   |
| 10           | How does <b>encryption</b> help in data security?                                   | BTL-2           | Understand        | CO1                   |
| 11           | Describe the role of <b>threat modeling</b> in identifying attack vectors.          | BTL-2           | Understand        | CO1                   |
| 12           | Explain <b>probabilistic reasoning</b> with respect to analyzing attack likelihood. | BTL-2           | Understand        | CO1                   |
| 13           | List any two <b>data security mitigation techniques</b> .                           | BTL-1           | Remember          | CO1                   |
| 14           | What is the purpose of <b>access control mechanisms</b> ?                           | BTL-1           | Remember          | CO1                   |
| 15           | Define <b>firewall</b> and its use in network security.                             | BTL-1           | Remember          | CO1                   |
| 16           | What is a <b>hashing algorithm</b> ?                                                | BTL-1           | Remember          | CO1                   |
| 17           | What is the difference between <b>symmetric</b> and <b>asymmetric encryption</b> .  | BTL-2           | Understand        | CO1                   |
| 18           | How does <b>multi-factor authentication (MFA)</b> enhance data security?            | BTL-2           | Understand        | CO1                   |

|    |                                                                            |       |            |     |
|----|----------------------------------------------------------------------------|-------|------------|-----|
| 19 | Discuss how <b>regular software updates</b> help mitigate vulnerabilities. | BTL-2 | Understand | CO1 |
| 20 | Describe how <b>intrusion detection systems (IDS)</b> work.                | BTL-2 | Understand | CO1 |
| 21 | What is <b>Privacy-Preserving Machine Learning (PPML)</b> ?                | BTL-1 | Remember   | CO1 |
| 22 | Name any two <b>open-source libraries</b> used for PPML.                   | BTL-1 | Remember   | CO1 |
| 23 | What is <b>differential privacy</b> ?                                      | BTL-1 | Remember   | CO1 |
| 24 | Define <b>federated learning</b> in ML.                                    | BTL-1 | Remember   | CO1 |

## UNIT - I: PART – B

CO1

| Q. No | Question                                                                                                                                                                | Mark | BT Level | Competence | Course Outcome |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|------------|----------------|
| 1     | Illustrate various <b>common attack vectors</b> (e.g., phishing, malware, MITM) with suitable examples. Design a simple attack scenario involving two of these vectors. | 16   | BTL-3    | Apply      | CO1            |
| 2     | Analyze the working of a <b>SQL Injection attack</b> and suggest a mitigation strategy by examining its vulnerabilities.                                                | 16   | BTL-4    | Analyze    | CO1            |
| 3     | Compare <b>phishing and spear-phishing</b> attacks with real-world examples. What makes spear-phishing harder to detect?                                                | 16   | BTL-4    | Analyze    | CO1            |
| 4     | Evaluate the impact of Advanced Persistent Threats (APTs) on enterprise security. Justify your answer with at least one case study.                                     | 16   | BTL-5    | Evaluate   | CO1            |
| 5     | Apply the concept of probabilistic reasoning to estimate the likelihood of a ransomware attack on a banking system. Use basic probabilities to model and explain.       | 16   | BTL-3    | Apply      | CO1            |
| 6     | Analyze the relationship between <b>data confidentiality, integrity, and availability (CIA)</b> with respect to data breach incidents.                                  | 16   | BTL-4    | Analyze    | CO1            |
| 7     | Examine how <b>attack trees</b> and <b>Bayesian models</b> help in understanding and predicting cyber attacks.                                                          | 16   | BTL-4    | Analyze    | CO1            |
| 8     | Critically evaluate the use of probabilistic reasoning models in cybersecurity risk assessment. What are the limitations?                                               | 16   | BTL-5    | Evaluate   | CO1            |
| 9     | Design a basic Bayesian Network model to analyze a security incident involving multiple attack paths (e.g., phishing, malware, insider threat).                         | 16   | BTL-3    | Apply      | CO1            |
| 10    | Implement a data security plan for an educational institution. Apply techniques like access control, encryption, and backup planning.                                   | 16   | BTL-3    | Apply      | CO1            |
| 11    | Analyze and compare the effectiveness of <b>firewalls, IDS, and antivirus software</b> in protecting against external threats.                                          | 16   | BTL-4    | Analyze    | CO1            |
| 12    | Differentiate <b>preventive, detective, and corrective</b> security controls using real-time examples.                                                                  | 16   | BTL-4    | Analyze    | CO1            |

|    |                                                                                                                                                                              |    |       |          |     |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-----|
| 13 | Evaluate the effectiveness of multi-layered security architecture (defense-in-depth) in enterprise systems.                                                                  | 16 | BTL-5 | Evaluate | CO1 |
| 14 | Evaluate the benefits and trade-offs of differential privacy vs. federated learning in real-time ML applications.                                                            | 16 | BTL-5 | Evaluate | CO1 |
| 15 | Analyze the privacy risks involved in ML-based healthcare applications. What can go wrong if privacy isn't preserved?                                                        | 16 | BTL-4 | Analyze  | CO1 |
| 16 | Design a privacy-preserving ML pipeline using federated learning and differential privacy techniques. Include tools and libraries used.                                      | 16 | BTL-6 | Create   | CO1 |
| 17 | Create an architecture for a privacy-preserving data analytics platform for a retail company. Include data minimization, anonymization, and compliance aspects (e.g., GDPR). | 16 | BTL-6 | Create   | CO1 |

## UNIT II ENCRYPTED COMPUTATION

Encrypted computation – Types of encrypted computation: Secure Multi-party computation – Homomorphic encryption. Real-world encrypted computation: Private set intersection – Private join and compute – Secure Aggregation – Encrypted Machine Learning. PSI and Moose.

### UNIT II PART – A

| Q. No | Questions                                                                                  | BT Level | Competence | Course Outcome |
|-------|--------------------------------------------------------------------------------------------|----------|------------|----------------|
| 1     | Define <b>encrypted computation</b> .                                                      | BTL-1    | Remember   | CO2            |
| 2     | What is <b>Secure Multi-party Computation (SMPC)</b> ?                                     | BTL-1    | Remember   | CO2            |
| 3     | Define <b>homomorphic encryption</b> .                                                     | BTL-1    | Remember   | CO2            |
| 4     | Name any two types of <b>homomorphic encryption</b> schemes.                               | BTL-1    | Remember   | CO2            |
| 5     | List any two <b>applications</b> of encrypted computation.                                 | BTL-1    | Remember   | CO2            |
| 6     | What do you mean by <b>semi-honest adversary</b> in SMPC?                                  | BTL-1    | Remember   | CO2            |
| 7     | Name two cryptographic techniques used in <b>SMPC</b> .                                    | BTL-1    | Remember   | CO2            |
| 8     | State the difference between <b>partially</b> and <b>fully homomorphic encryption</b> .    | BTL-1    | Remember   | CO2            |
| 9     | Explain how <b>homomorphic encryption</b> allows computation on encrypted data.            | BTL-2    | Understand | CO2            |
| 10    | Compare <b>SMPC</b> and <b>homomorphic encryption</b> in terms of privacy and computation. | BTL-2    | Understand | CO2            |
| 11    | Illustrate the need for <b>encrypted computation</b> in cloud environments.                | BTL-2    | Understand | CO2            |
| 12    | Explain the term " <b>computation over encrypted data</b> " with an example.               | BTL-2    | Understand | CO2            |
| 13    | What is <b>Private Set Intersection (PSI)</b> ?                                            | BTL-1    | Remember   | CO2            |
| 14    | Define <b>Private Join and Compute (PJC)</b> .                                             | BTL-1    | Remember   | CO2            |
| 15    | What is <b>Secure Aggregation</b> ?                                                        | BTL-1    | Remember   | CO2            |
| 16    | Mention one <b>real-world application</b> of PSI.                                          | BTL-1    | Remember   | CO2            |
| 17    | What does the term <b>privacy-preserving aggregation</b> mean?                             | BTL-1    | Remember   | CO2            |

|    |                                                                                   |       |            |     |
|----|-----------------------------------------------------------------------------------|-------|------------|-----|
| 18 | Name any two organizations using Secure Aggregation in production.                | BTL-1 | Remember   | CO2 |
| 19 | Describe the process of <b>Private Set Intersection</b> with an example scenario. | BTL-2 | Understand | CO2 |
| 20 | Explain how <b>Secure Aggregation</b> ensures data privacy in federated learning. | BTL-2 | Understand | CO2 |
| 21 | Compare <b>Private Join and Compute</b> with PSI.                                 | BTL-2 | Understand | CO2 |
| 22 | +Discuss the use of <b>encrypted machine learning</b> in healthcare or finance.   | BTL-2 | Understand | CO2 |
| 23 | What is <b>Moose</b> in the context of encrypted computation?                     | BTL-2 | Understand | CO2 |
| 24 | List one difference between <b>PSI</b> and <b>Moose</b> .                         | BTL-1 | Remember   | CO2 |

| UNIT II PART – B |                                                                                                                                                                |      |          |            |                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|------------|----------------|
| Q. No            | Question                                                                                                                                                       | Mark | BT Level | Competence | Course Outcome |
| 1                | <b>Design a scenario</b> in which encrypted computation is necessary. Show how either <b>homomorphic encryption</b> or <b>SMPC</b> can be applied to solve it. | 16   | BTL-3    | Apply      | CO2            |
| 2                | Apply <b>partially homomorphic encryption</b> to compute the sum of encrypted data without decrypting it. Explain step by step.                                | 16   | BTL-3    | Apply      | CO2            |
| 3                | Construct a real-world application where <b>SMPC</b> is used to compute a function collaboratively without revealing individual inputs.                        | 16   | BTL-3    | Apply      | CO2            |
| 4                | Analyze the trade-offs between <b>homomorphic encryption</b> and <b>secure multi-party computation</b> in terms of computational cost and privacy.             | 16   | BTL-4    | Analyze    | CO2            |
| 5                | Compare and contrast <b>fully, partially, and somewhat</b> homomorphic encryption with examples.                                                               | 16   | BTL-4    | Analyze    | CO2            |
| 6                | Analyze the limitations of using <b>encrypted computation</b> in large-scale cloud data processing.                                                            | 16   | BTL-4    | Analyze    | CO2            |
| 7                | Explain how the choice of encrypted computation technique depends on the <b>threat model</b> (e.g., honest-but-curious vs malicious adversaries).              | 16   | BTL-4    | Analyze    | CO2            |
| 8                | Evaluate the effectiveness of <b>SMPC protocols</b> (like Yao's Garbled Circuits or GMW) in financial use cases such as joint fraud detection.                 | 16   | BTL-5    | Evaluate   | CO2            |
| 9                | Assess the suitability of <b>homomorphic encryption</b> in a healthcare data analytics system. What are the privacy and performance implications?              | 16   | BTL-5    | Evaluate   | CO2            |
| 10               | Compare the <b>usability, performance, and security guarantees</b> of SMPC and homomorphic encryption in the context of secure elections.                      | 16   | BTL-5    | Evaluate   | CO2            |
| 11               | Analyze the challenges in implementing <b>encrypted machine learning</b> and how real-world frameworks attempt to overcome them.                               | 16   | BTL-4    | Analyze    | CO2            |
| 12               | Discuss how <b>Private Join and Compute</b> balances computation cost                                                                                          | 16   | BTL-4    | Analyze    | CO2            |

|    |                                                                                                                                                                       |    |       |          |     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-----|
|    | and data confidentiality when working across datasets.                                                                                                                |    |       |          |     |
| 13 | Apply <b>Private Set Intersection (PSI)</b> in a contact tracing scenario where privacy of individuals is preserved.                                                  | 16 | BTL-3 | Apply    | CO2 |
| 14 | Demonstrate the use of <b>Private Join and Compute</b> in ad conversion tracking with encrypted user identifiers.                                                     | 16 | BTL-3 | Apply    | CO2 |
| 15 | Illustrate how <b>Secure Aggregation</b> works in federated learning. Show how client updates are encrypted and aggregated.                                           | 16 | BTL-3 | Apply    | CO2 |
| 16 | Evaluate the impact of using <b>encrypted computation</b> techniques on model accuracy and system performance in ML pipelines.                                        | 16 | BTL-5 | Evaluate | CO2 |
| 17 | Design a secure application using <b>Moose</b> and <b>PSI</b> to allow hospitals to compute common patient statistics across institutions without revealing raw data. | 16 | BTL-6 | Create   | CO2 |

### UNIT – III DATA GOVERNANCE AND PRIVACY APPROACHES

Data Governance – Identifying sensitive data – Documenting data for use - Basic Privacy – Anonymization – Differential privacy – Privacy loss – Differential privacy with Laplace mechanism – Gaussian noise for differential privacy – Sensitivity and Privacy units – kAnonymity – Building Privacy into Data Pipelines.

| Q. No                    | Questions                                                                                                       | BT Level | Competence | Course Outcome |
|--------------------------|-----------------------------------------------------------------------------------------------------------------|----------|------------|----------------|
| <b>UNIT III PART – A</b> |                                                                                                                 |          |            |                |
| 1                        | Define <b>data governance</b> .                                                                                 | BTL-1    | Remember   | CO3            |
| 2                        | What is meant by <b>sensitive data</b> ?                                                                        | BTL-1    | Remember   | CO3            |
| 3                        | List any two examples of <b>personally identifiable information (PII)</b> .                                     | BTL-1    | Remember   | CO3            |
| 4                        | What do you mean by <b>data documentation</b> ?                                                                 | BTL-1    | Remember   | CO3            |
| 5                        | Define the term <b>metadata</b> in the context of data governance.                                              | BTL-1    | Remember   | CO3            |
| 6                        | State two key objectives of <b>data governance policies</b> .                                                   | BTL-1    | Remember   | CO3            |
| 7                        | Explain why it is important to <b>identify and label sensitive data</b> .                                       | BTL-2    | Understand | CO3            |
| 8                        | Describe how <b>data documentation</b> supports transparency in data usage.                                     | BTL-2    | Understand | CO3            |
| 9                        | Differentiate between <b>sensitive</b> and <b>non-sensitive data</b> with examples.                             | BTL-2    | Understand | CO3            |
| 10                       | Explain the relationship between <b>data governance</b> and <b>compliance requirements</b> (e.g., GDPR, HIPAA). | BTL-2    | Understand | CO3            |
| 11                       | What is <b>basic data privacy</b> ?                                                                             | BTL-1    | Remember   | CO3            |
| 12                       | Define <b>anonymization</b> .                                                                                   | BTL-1    | Remember   | CO3            |
| 13                       | What is <b>k-anonymity</b> ?                                                                                    | BTL-1    | Remember   | CO3            |
| 14                       | List two techniques used for <b>anonymization</b> of data.                                                      | BTL-1    | Remember   | CO3            |
| 15                       | What is a <b>quasi-identifier</b> ?                                                                             | BTL-1    | Remember   | CO3            |
| 16                       | Define <b>re-identification risk</b> .                                                                          | BTL-1    | Remember   | CO3            |
| 17                       | Explain the concept of <b>k-anonymity</b> with a simple example.                                                | BTL-2    | Understand | CO3            |
| 18                       | Discuss the limitations of <b>basic anonymization techniques</b> .                                              | BTL-2    | Understand | CO3            |
| 19                       | Differentiate between <b>anonymization</b> and <b>pseudonymization</b> .                                        | BTL-2    | Understand | CO3            |
| 20                       | Why is <b>k-anonymity</b> considered insufficient for high-risk datasets?                                       | BTL-2    | Understand | CO3            |
| 21                       | Define differential privacy.                                                                                    | BTL-1    | Remember   | CO3            |

|    |                                                                    |       |          |     |
|----|--------------------------------------------------------------------|-------|----------|-----|
| 22 | What is the <b>Laplace mechanism</b> in differential privacy?      | BTL-1 | Remember | CO3 |
| 23 | What is meant by <b>sensitivity</b> in differential privacy?       | BTL-1 | Remember | CO3 |
| 24 | What is the role of <b>Gaussian noise</b> in privacy preservation? | BTL-1 | Remember | CO3 |

| UNIT III PART – B |                                                                                                                                                                                          |      |          |            |                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|------------|----------------|
| Q. No             | Question                                                                                                                                                                                 | Mark | BT Level | Competence | Course Outcome |
| 1                 | Apply the principles of <b>data governance</b> to design a data access policy for a healthcare institution.                                                                              | 16   | BTL-3    | Apply      | CO3            |
| 2                 | Demonstrate how to <b>identify sensitive data</b> in a banking dataset using data classification techniques.                                                                             | 16   | BTL-3    | Apply      | CO3            |
| 3                 | Create a data documentation plan for a dataset used in a <b>machine learning project</b> , including metadata and data lineage.                                                          | 16   | BTL-3    | Apply      | CO3            |
| 4                 | Analyze the <b>challenges in identifying sensitive data</b> in a large enterprise environment with structured and unstructured data.                                                     | 16   | BTL-4    | Analyze    | CO3            |
| 5                 | Compare and contrast different <b>data documentation approaches</b> and explain their impact on privacy compliance and auditability.                                                     | 16   | BTL-4    | Analyze    | CO3            |
| 6                 | Apply <b>anonymization techniques</b> to transform a sample dataset while retaining its analytical value.                                                                                | 16   | BTL-3    | Apply      | CO3            |
| 7                 | Use a real-life example to explain how <b>k-anonymity</b> can protect user identities in a public dataset.                                                                               | 16   | BTL-3    | Apply      | CO3            |
| 8                 | Analyze the risks of <b>re-identification</b> in anonymized datasets and explain how <b>quasi-identifiers</b> contribute to the problem.                                                 | 16   | BTL-4    | Analyze    | CO3            |
| 9                 | Compare <b>k-anonymity, l-diversity, and t-closeness</b> . In which scenarios is each more effective?                                                                                    | 16   | BTL-4    | Analyze    | CO3            |
| 10                | Evaluate the effectiveness of <b>traditional anonymization techniques</b> versus <b>modern privacy-preserving methods</b> (like differential privacy) in the context of e-commerce data. | 16   | BTL-5    | Evaluate   | CO3            |
| 11                | Use a numerical example to demonstrate the working of <b>Gaussian noise</b> in differential privacy.                                                                                     | 16   | BTL-3    | Apply      | CO3            |
| 12                | Analyze how <b>privacy loss</b> is calculated and managed in differential privacy. What does the privacy budget represent?                                                               | 16   | BTL-4    | Analyze    | CO3            |
| 13                | Differentiate between <b>Laplace mechanism and Gaussian mechanism</b> with their mathematical formulation and use cases.                                                                 | 16   | BTL-4    | Analyze    | CO3            |
| 14                | Design a <b>data pipeline</b> that incorporates privacy-preserving steps such as encryption, access control, and anonymization.                                                          | 16   | BTL-3    | Apply      | CO3            |
| 15                | Implement a privacy-aware data flow in a <b>real-time analytics pipeline</b> using tools like Apache Kafka or Spark.                                                                     | 16   | BTL-3    | Apply      | CO3            |
| 16                | Propose an architecture to embed <b>differential privacy mechanisms</b> directly into an ML training pipeline.                                                                           | 16   | BTL-6    | Create     | CO3            |
| 17                | Create a complete privacy governance framework for a company that handles user behavioral data for targeted advertising.                                                                 | 16   | BTL-6    | Create     | CO3            |

#### UNIT – IV FEDERATED LEARNING AND DATA SCIENCE

Distributed data – Distributed Optimization - Federated learning – Architecting federated systems – Open-source federated libraries – Federated data science

| Q. No                   | Questions                                                                                                     | BT Level | Competence | Course Outcome |
|-------------------------|---------------------------------------------------------------------------------------------------------------|----------|------------|----------------|
| <b>UNIT IV PART – A</b> |                                                                                                               |          |            |                |
| 1                       | Define <b>distributed data</b> .                                                                              | BTL-1    | Remember   | CO4            |
| 2                       | What is <b>data partitioning</b> in distributed systems?                                                      | BTL-1    | Remember   | CO4            |
| 3                       | List two examples of <b>distributed data storage systems</b> .                                                | BTL-1    | Remember   | CO4            |
| 4                       | What is meant by <b>data locality</b> ?                                                                       | BTL-1    | Remember   | CO4            |
| 5                       | Define <b>distributed optimization</b> .                                                                      | BTL-1    | Remember   | CO4            |
| 6                       | Name any two techniques used in <b>distributed optimization</b> .                                             | BTL-1    | Remember   | CO4            |
| 7                       | What is a <b>parameter server</b> in distributed learning?                                                    | BTL-1    | Remember   | CO4            |
| 8                       | Mention any two challenges in handling distributed data.                                                      | BTL-1    | Remember   | CO4            |
| 9                       | What is <b>federated learning</b> ?                                                                           | BTL-1    | Remember   | CO4            |
| 10                      | List two advantages of federated learning.                                                                    | BTL-1    | Remember   | CO4            |
| 11                      | What is a <b>federated averaging algorithm (FedAvg)</b> ?                                                     | BTL-1    | Remember   | CO4            |
| 12                      | Define <b>client drift</b> in federated learning.                                                             | BTL-1    | Remember   | CO4            |
| 13                      | Name any two devices where federated learning is commonly used.                                               | BTL-1    | Remember   | CO4            |
| 14                      | What is <b>model aggregation</b> in federated systems?                                                        | BTL-1    | Remember   | CO4            |
| 15                      | Explain the difference between <b>federated learning</b> and <b>centralized learning</b> .                    | BTL-2    | Understand | CO4            |
| 16                      | Describe the concept of <b>on-device learning</b> in federated environments.                                  | BTL-2    | Understand | CO4            |
| 17                      | Discuss how <b>privacy</b> is maintained in federated learning.                                               | BTL-2    | Understand | CO4            |
| 18                      | Explain how <b>federated learning supports personalization</b> of models.                                     | BTL-2    | Understand | CO4            |
| 19                      | Identify key components of a <b>federated learning architecture</b> (e.g., client, server, aggregator).       | BTL-2    | Understand | CO4            |
| 20                      | Describe the challenges of <b>communication efficiency</b> in federated systems.                              | BTL-2    | Understand | CO4            |
| 21                      | <b>Discuss</b> any two key features of PySyft and how they support federated learning.                        | BTL-2    | Understand | CO4            |
| 22                      | <b>Explain</b> the concept of federated data science in your own words.                                       | BTL-2    | Understand | CO4            |
| 23                      | <b>Describe</b> two real-world applications of federated data science and how they benefit from data privacy. | BTL-2    | Understand | CO4            |
| 24                      | <b>Interpret</b> two commonly used metrics to evaluate performance in federated learning environments.        | BTL-2    | Understand | CO4            |

| <b>UNIT IV PART – B</b> |                                                                                                                                        |      |          |            |                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------|----------|------------|----------------|
| Q. No                   | Question                                                                                                                               | Mark | BT Level | Competence | Course Outcome |
| 1                       | Apply a <b>distributed data architecture</b> for a multinational company managing real-time analytics across geolocations.             | 16   | BTL-3    | Apply      | CO4            |
| 2                       | Demonstrate how <b>gradient descent</b> can be adapted to work in a distributed environment.                                           | 16   | BTL-3    | Apply      | CO4            |
| 3                       | Illustrate the role of <b>parameter servers</b> in distributed optimization with a case study from a large-scale ML training scenario. | 16   | BTL-3    | Apply      | CO4            |
| 4                       | Construct a federated learning scenario involving mobile devices                                                                       | 16   | BTL-3    | Apply      | CO4            |

|    |                                                                                                                                                                       |    |       |          |     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----------|-----|
|    | using FedAvg and explain the step-by-step flow.                                                                                                                       |    |       |          |     |
| 5  | Analyze the impact of <b>data partitioning strategies</b> (horizontal vs. vertical) on performance and fault tolerance.                                               | 16 | BTL-4 | Analyze  | CO4 |
| 6  | Compare and contrast <b>synchronous vs. asynchronous distributed optimization</b> with respect to convergence and scalability.                                        | 16 | BTL-4 | Analyze  | CO4 |
| 7  | Examine the trade-offs between <b>data consistency and scalability</b> in distributed data systems.                                                                   | 16 | BTL-4 | Analyze  | CO4 |
| 8  | Apply federated learning to a healthcare application involving multiple hospitals with patient data privacy constraints.                                              | 16 | BTL-3 | Apply    | CO4 |
| 9  | Analyze the problem of <b>client drift</b> and its impact on model convergence in non-IID federated data.                                                             | 16 | BTL-4 | Analyze  | CO4 |
| 10 | Compare <b>federated learning</b> and <b>traditional centralized learning</b> in terms of privacy, latency, and model performance.                                    | 16 | BTL-4 | Analyze  | CO4 |
| 11 | Discuss the challenges of <b>handling stragglers and dropped clients</b> in federated learning environments                                                           | 16 | BTL-4 | Analyze  | CO4 |
| 12 | Break down the stages of a federated learning system pipeline, highlighting where architectural optimizations can be made.                                            | 16 | BTL-4 | Analyze  | CO4 |
| 13 | Design a high-level architecture for a federated system used in real-time predictive maintenance in manufacturing.                                                    | 16 | BTL-3 | Apply    | CO4 |
| 14 | Assess the benefits and limitations of using <b>open-source libraries</b> for deploying federated learning at scale.                                                  | 16 | BTL-5 | Evaluate | CO4 |
| 15 | Compare the <b>deployment flexibility, scalability, and community support</b> among federated libraries like FATE, OpenFL, and Flower.                                | 16 | BTL-5 | Evaluate | CO4 |
| 16 | Propose a federated data science workflow for a <b>collaborative fraud detection system</b> across multiple financial institutions without sharing raw data.          | 16 | BTL-6 | Create   | CO4 |
| 17 | Design a complete <b>federated learning architecture</b> with edge devices, model server, aggregator, and privacy-preserving techniques for a smart city application. | 16 | BTL-6 | Create   | CO4 |

| UNIT – V LEGALITY OF PRIVACY                                                                                                                                                                                                          |                                                                                      |          |            |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|------------|----------------|
| GDPR – CCPA – HIPAA - LGPD - PIPL- Internal policies and contracts – Adhering to contract agreements and law – Interpreting Data protection regulations – Data governance 2.0 - Indian Data Protection Framework - Use case analysis. |                                                                                      |          |            |                |
| Q. No                                                                                                                                                                                                                                 | Questions                                                                            | BT Level | Competence | Course Outcome |
| UNIT V PART – A                                                                                                                                                                                                                       |                                                                                      |          |            |                |
| 1                                                                                                                                                                                                                                     | Define <b>GDPR</b> .                                                                 | BTL-1    | Remember   | CO5            |
| 2                                                                                                                                                                                                                                     | List two key rights provided to individuals under GDPR.                              | BTL-1    | Remember   | CO5            |
| 3                                                                                                                                                                                                                                     | What is the purpose of a <b>Data Protection Impact Assessment (DPIA)</b> under GDPR? | BTL-1    | Remember   | CO5            |
| 4                                                                                                                                                                                                                                     | Describe the concept of <b>consent</b> under GDPR.                                   | BTL-1    | Remember   | CO5            |
| 5                                                                                                                                                                                                                                     | Explain the role of a <b>Data Protection Officer (DPO)</b> as per GDPR.              | BTL-1    | Remember   | CO5            |

|    |                                                                                             |       |            |     |
|----|---------------------------------------------------------------------------------------------|-------|------------|-----|
| 6  | Define <b>CCPA</b> and mention one of its main objectives.                                  | BTL-1 | Remember   | CO5 |
| 7  | What are two rights given to consumers under the CCPA?                                      | BTL-1 | Remember   | CO5 |
| 8  | Differentiate between <b>GDPR</b> and <b>CCPA</b> in terms of scope and applicability.      | BTL-1 | Remember   | CO5 |
| 9  | What is the primary goal of <b>HIPAA</b> ?                                                  | BTL-1 | Remember   | CO5 |
| 10 | List two types of entities covered under HIPAA.                                             | BTL-1 | Remember   | CO5 |
| 11 | Define <b>Protected Health Information (PHI)</b> .                                          | BTL-1 | Remember   | CO5 |
| 12 | What is <b>LGPD</b> ?                                                                       | BTL-1 | Remember   | CO5 |
| 13 | Mention two similarities between <b>LGPD</b> and <b>GDPR</b> .                              | BTL-1 | Remember   | CO5 |
| 14 | Describe the term <b>data subject rights</b> under LGPD.                                    | BTL-2 | Understand | CO5 |
| 15 | What is <b>PIPL</b> ?                                                                       | BTL-2 | Understand | CO5 |
| 16 | List two key principles of data protection under PIPL.                                      | BTL-2 | Understand | CO5 |
| 17 | Explain how <b>user consent</b> is handled in PIPL.                                         | BTL-2 | Understand | CO5 |
| 18 | Describe the role of <b>data localization</b> under PIPL.                                   | BTL-2 | Understand | CO5 |
| 19 | Compare <b>PIPL</b> and <b>GDPR</b> in terms of enforcement.                                | BTL-2 | Understand | CO5 |
| 20 | Define <b>internal data policy</b> in the context of data protection.                       | BTL-2 | Understand | CO5 |
| 21 | Mention two reasons why <b>contract compliance</b> is important in data privacy.            | BTL-2 | Understand | CO5 |
| 22 | Explain the need for <b>interpreting data protection regulations</b> accurately.            | BTL-2 | Understand | CO5 |
| 23 | Describe how organizations can <b>adhere to data sharing contracts</b> .                    | BTL-2 | Understand | CO5 |
| 24 | What is <b>Data Governance 2.0</b> and how does it differ from traditional data governance? | BTL-2 | Understand | CO5 |

| UNIT V PART – B |                                                                                                                                                 |      |          |            |                |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|------------|----------------|
| Q. No           | Question                                                                                                                                        | Mark | BT Level | Competence | Course Outcome |
| 1               | Apply the key principles of <b>GDPR</b> to design a privacy-compliant data handling system for a European e-commerce platform.                  | 16   | BTL-3    | Apply      | CO5            |
| 2               | Demonstrate how a <b>healthcare application</b> should be designed to comply with <b>HIPAA</b> regulations.                                     | 16   | BTL-3    | Apply      | CO5            |
| 3               | Illustrate how a company based in California must handle user data to comply with <b>CCPA</b> .                                                 | 16   | BTL-3    | Apply      | CO5            |
| 4               | Implement the <b>Indian Data Protection Bill</b> principles in the development of a local fintech app.                                          | 16   | BTL-3    | Apply      | CO5            |
| 5               | Show how <b>internal privacy policies and contracts</b> can be aligned with global data protection laws in a multinational organization.        | 16   | BTL-3    | Apply      | CO5            |
| 6               | Analyze the similarities and differences between <b>GDPR</b> , <b>CCPA</b> , and <b>PIPL</b> in terms of consent, user rights, and enforcement. | 16   | BTL-4    | Analyze    | CO5            |
| 7               | Examine the impact of <b>LGPD</b> on Brazilian startups handling personal data of EU and non-EU citizens.                                       | 16   | BTL-4    | Analyze    | CO5            |
| 8               | Compare <b>HIPAA</b> and <b>GDPR</b> in the context of healthcare data privacy and cross-border data flow.                                      | 16   | BTL-4    | Analyze    | CO5            |
| 9               | Analyze a <b>real-world data breach</b> case and evaluate the failure of adherence to data privacy laws.                                        | 16   | BTL-4    | Analyze    | CO5            |
| 10              | Break down the key elements of <b>Data Governance 2.0</b> and how it improves upon traditional governance practices.                            | 16   | BTL-4    | Analyze    | CO5            |

|    |                                                                                                                                                            |    |       |          |     |
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| 11 | Analyze the key components of Data Governance 2.0 and explain how each can contribute to improved transparency and compliance in a government data portal. | 16 | BTL-4 | Analyze  | CO5 |
| 12 | Design a data privacy framework that integrates GDPR, HIPAA, and Indian data protection laws for a multinational healthtech company.                       | 16 | BTL-6 | Create   | CO5 |
| 13 | <b>Demonstrate</b> how a unified policy structure can be implemented to manage internal contracts and data sharing in compliance with local laws.          | 16 | BTL-3 | Apply    | CO5 |
| 14 | <b>Illustrate</b> how a financial services app operating in California and Brazil can meet the data privacy requirements of both CCPA and LGPD.            | 16 | BTL-3 | Apply    | CO5 |
| 15 | Justify whether the <b>Personal Data Protection Bill (India)</b> sufficiently addresses privacy concerns for citizens and tech companies.                  | 16 | BTL-5 | Evaluate | CO5 |
| 16 | Critically assess the role of <b>Data Protection Officers (DPOs)</b> under different privacy regulations like GDPR and PIPL.                               | 16 | BTL-5 | Evaluate | CO5 |
| 17 | Create a compliance checklist for an Indian startup handling EU, US, and Chinese citizen data.                                                             | 16 | BTL-6 | Create   | CO5 |